Maxwell Equations

before Maxwell…..

$$\vec{\nabla}\cdot\vec{\mathbf{E}}=\frac{\rho}{\varepsilon_0}$$ **Gauss's law**

$$\vec{\nabla}\cdot\vec{\mathbf{B}}=0$$ **No name**

$$\vec{\nabla}\times\vec{\mathbf{E}}=-\frac{\partial\vec{\mathbf{B}}}{\partial t}$$ **Faraday's Law**

$$\vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\vec{\mathbf{J}}$$ **Ampere's Law**

$$\vec{\nabla}\cdot\vec{\mathbf{J}}+\frac{\partial\rho}{\partial t}=0$$

are they mathematically and physics-wise consistent?

$$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{\mathbf{E}})=\vec{\nabla}\cdot\left(-\frac{\partial\vec{\mathbf{B}}}{\partial t}\right)=-\frac{\partial(\vec{\nabla}\cdot\vec{\mathbf{B}})}{\partial t}=0\ \checkmark$$

$$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{\mathbf{B}})=\vec{\nabla}\cdot(\mu_0\vec{\mathbf{J}})=\mu_0(\vec{\nabla}\cdot\vec{\mathbf{J}})=-\mu_0\frac{\partial\rho}{\partial t}$$

Always zero

Zero only in steady state condition

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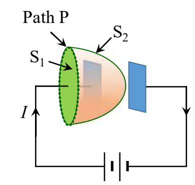
are they mathematically and physics-wise consistent?

$$\oint\vec{\mathbf{B}}\cdot d\vec{\mathbf{l}}=\mu_0 I_{en}=\mu_0 I \quad \text{(through } S_1\text{)}$$

$$=0 \quad \text{(through } S_2\text{)}$$



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Maxwell Equations

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$$\vec{\nabla}\cdot\vec{\mathbf{J}}+\frac{\partial\rho}{\partial t}=0$$

…how Maxwell fixed Ampere's law?

$$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{\mathbf{B}})=\vec{\nabla}\cdot(\mu_0\vec{\mathbf{J}})=\mu_0(\vec{\nabla}\cdot\vec{\mathbf{J}})$$

$$\vec{\nabla}\cdot\vec{\mathbf{J}}=-\frac{\partial\rho}{\partial t}=-\frac{\partial}{\partial t}(\varepsilon_0\vec{\nabla}\cdot\vec{\mathbf{E}})=-\vec{\nabla}\cdot\left(\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)$$

$$\Rightarrow\ \vec{\nabla}\cdot\left(\vec{\mathbf{J}}+\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)=0$$

$$\boxed{\vec{\mathbf{J}}_d=\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}}$$

Displacement current

$$\boxed{\vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\vec{\mathbf{J}}+\varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t}}$$

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Maxwell Equations

…how Maxwell fixed Ampere's law?

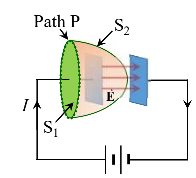
$$\vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\left(\vec{\mathbf{J}}+\vec{\mathbf{J}}_d\right) \qquad \vec{\mathbf{J}}_d=\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$ **Displacement current**

$$\oint\vec{\mathbf{B}}\cdot d\vec{\mathbf{l}}=\mu_0 I_{en}+\varepsilon_0\mu_0\int\left(\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)\cdot d\vec{\mathbf{a}}$$

$$E=\frac{\sigma}{\varepsilon_0}=\frac{1}{\varepsilon_0}\frac{Q}{A}\ \Rightarrow\ \frac{\partial E}{\partial t}=\frac{1}{\varepsilon_0 A}\frac{dQ}{dt}=\frac{I}{\varepsilon_0 A}$$

For S_1: $E=0;\ I_{en}=I\ \Rightarrow\ \oint\vec{\mathbf{B}}\cdot d\vec{\mathbf{l}}=\mu_0 I+0$

For S_2: $I_{en}=0;\ \int\left(\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)\cdot d\vec{\mathbf{a}}=\frac{I}{\varepsilon_0\cancel{A}}\cancel{A}\ \Rightarrow\ \oint\vec{\mathbf{B}}\cdot d\vec{\mathbf{l}}=0+\cancel{\varepsilon_0}\mu_0\frac{I}{\cancel{\varepsilon_0}}$



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Maxwell Equations

Maxwell contribution and impact?

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0}$$

$$\vec{\nabla}\cdot\vec{\mathbf{B}} = 0$$

$$\vec{\nabla}\times\vec{\mathbf{E}} = -\frac{\partial\vec{\mathbf{B}}}{\partial t}$$

$$\vec{\nabla}\times\vec{\mathbf{B}} = \mu_0\vec{\mathbf{J}} + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

- Seemingly minor change in the field equation
  ...fundamental change in the theory of Electromagnetics!
- Idea of displacement current vastly extends our physical conceptions of the world
  ...from the point of view of physics, light that reaches to us is nothing but displacement current!
- Displacement current is of importance only
  - if $\vec{\mathbf{J}} = 0$ (propagation of light in dielectric)
  - in the case of high frequencies when $\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t} \propto \omega$
- Experimental proof only in 1888 by Heinrich Hertz

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Maxwell Equations

Maxwell contribution and impact?

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0}$$

$$\vec{\nabla}\cdot\vec{\mathbf{B}} = 0$$

$$\vec{\nabla}\times\vec{\mathbf{E}} = -\frac{\partial\vec{\mathbf{B}}}{\partial t}$$

$$\vec{\nabla}\times\vec{\mathbf{B}} = \mu_0\vec{\mathbf{J}} + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

$$\vec{\mathbf{F}}_{\text{tot}} = q\,[\vec{\mathbf{E}} + (\vec{\mathbf{v}}\times\vec{\mathbf{B}})]$$

- Most important implications of ***displacement current*** and ***Maxwell equations***:
  - Interdependence of Electric and magnetic field
    ...time varying electric field induces magnetic field!
  - Existence of electromagnetic (EM) waves
  - Finite speed of propagation of EM waves
  - Propagation in free space at the speed of light
  - Light itself an EM wave

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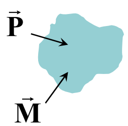
Maxwell Equations in Matter



$$\left.\begin{aligned}\rho_b &= -\vec{\nabla}\cdot\vec{\mathbf{P}}\\ \vec{\mathbf{J}}_b &= \vec{\nabla}\times\vec{\mathbf{M}}\end{aligned}\right\}\ \text{Static case}$$

Non-Static case???

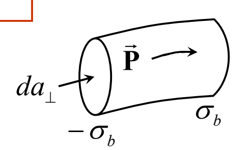
$$dI = \frac{\partial\sigma_b}{\partial t}da_\perp = \frac{\partial P}{\partial t}da_\perp \Rightarrow \vec{\mathbf{J}}_p = \frac{\partial\vec{\mathbf{P}}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{\mathbf{J}}_p = \vec{\nabla}\cdot\frac{\partial\vec{\mathbf{P}}}{\partial t} = \frac{\partial}{\partial t}(\vec{\nabla}\cdot\vec{\mathbf{P}}) = -\frac{\partial\rho_b}{\partial t} \Rightarrow \boxed{\vec{\nabla}\cdot\vec{\mathbf{J}} - \frac{\partial\rho_b}{\partial t} = 0}$$

Conservation of bound charges

- Total charge density: $\rho = \rho_f + \rho_b = \rho_f - \vec{\nabla}\cdot\vec{\mathbf{P}}$
- Total current density: $\vec{\mathbf{J}} = \vec{\mathbf{J}}_f + \vec{\mathbf{J}}_b + \vec{\mathbf{J}}_p$

$$= \vec{\mathbf{J}}_f + \vec{\nabla}\times\vec{\mathbf{M}} + \frac{\partial\vec{\mathbf{P}}}{\partial t}$$



$$\sigma_b = \vec{\mathbf{P}}\cdot\hat{\mathbf{n}}$$

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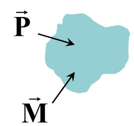
Maxwell Equations in Matter



- Total charge density: $\rho = \rho_f - \vec{\nabla}\cdot\vec{\mathbf{P}}$
- Total current density: $\vec{\mathbf{J}} = \vec{\mathbf{J}}_f + \vec{\nabla}\times\vec{\mathbf{M}} + \frac{\partial\vec{\mathbf{P}}}{\partial t}$

Gauss's law $\vec{\nabla}\cdot\vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0} = \frac{1}{\varepsilon_0}\left(\rho_f - \vec{\nabla}\cdot\vec{\mathbf{P}}\right) \quad \times\varepsilon_0$

$$\Rightarrow \varepsilon_0\vec{\nabla}\cdot\vec{\mathbf{E}} + \vec{\nabla}\cdot\vec{\mathbf{P}} = \rho_f \Rightarrow \vec{\nabla}\cdot(\varepsilon_0\vec{\mathbf{E}} + \vec{\mathbf{P}}) = \rho_f \Rightarrow \boxed{\vec{\nabla}\cdot\vec{\mathbf{D}} = \rho_f} \quad \boxed{\vec{\mathbf{D}} = \varepsilon_0\vec{\mathbf{E}} + \vec{\mathbf{P}}}$$

Ampere's law $\vec{\nabla}\times\vec{\mathbf{B}} = \mu_0\vec{\mathbf{J}} + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t} = \mu_0\left(\vec{\mathbf{J}}_f + \vec{\nabla}\times\vec{\mathbf{M}} + \frac{\partial\vec{\mathbf{P}}}{\partial t}\right) + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$

$$\vec{\nabla}\times\vec{\mathbf{B}} - \mu_0\vec{\nabla}\times\vec{\mathbf{M}} = \mu_0\vec{\mathbf{J}}_f + \mu_0\frac{\partial\vec{\mathbf{P}}}{\partial t} + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t} \quad \times\frac{1}{\mu_0}$$

$$\vec{\nabla}\times\left(\frac{1}{\mu_0}\vec{\mathbf{B}} - \vec{\mathbf{M}}\right) = \vec{\mathbf{J}}_f + \frac{\partial}{\partial t}(\vec{\mathbf{P}} + \varepsilon_0\vec{\mathbf{E}}) \Rightarrow \boxed{\vec{\nabla}\times\vec{\mathbf{H}} = \vec{\mathbf{J}}_f + \frac{\partial\vec{\mathbf{D}}}{\partial t}} \quad \boxed{\vec{\mathbf{H}} = \frac{1}{\mu_0}\vec{\mathbf{B}} - \vec{\mathbf{M}}}$$

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Maxwell Equations in Matter

$$\vec{\nabla}\cdot\vec{\mathbf{D}}=\rho_f \qquad \vec{\nabla}\cdot\vec{\mathbf{B}}=0 \qquad \vec{\nabla}\times\vec{\mathbf{E}}=-\frac{\partial\vec{\mathbf{B}}}{\partial t} \qquad \vec{\nabla}\times\vec{\mathbf{H}}=\vec{\mathbf{J}}_f+\frac{\partial\vec{\mathbf{D}}}{\partial t}$$

Displacement current

$$\vec{\mathbf{D}}=\varepsilon_0\vec{\mathbf{E}}+\vec{\mathbf{P}} \qquad \vec{\mathbf{H}}=\frac{\vec{\mathbf{B}}}{\mu_0}-\vec{\mathbf{M}}$$

$$\vec{\mathbf{D}}=\varepsilon\,\vec{\mathbf{E}} \qquad \vec{\mathbf{B}}=\mu\,\vec{\mathbf{H}}$$

$$\vec{\mathbf{P}}=\varepsilon_0\chi_e\vec{\mathbf{E}} \qquad \vec{\mathbf{M}}=\chi_m\,\vec{\mathbf{H}}$$

$$\varepsilon=\varepsilon_0(1+\chi_e) \qquad \mu=\mu_0(1+\chi_m)$$

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Boundary Conditions for LIH Medium

- $\hat{n}\cdot(\vec{D}_2-\vec{D}_1)=\sigma_f$ ➡ $\vec{D}_2^{\perp}-\vec{D}_1^{\perp}=\sigma_f$
- $\hat{n}\times(\vec{E}_2-\vec{E}_1)=0$ ➡ $\vec{E}_2^{II}=\vec{E}_1^{II}$
- $\hat{n}\cdot(\vec{B}_2-\vec{B}_1)=0$ ➡ $\vec{B}_2^{\perp}=\vec{B}_1^{\perp}$
- $\hat{n}\times(\vec{H}_2-\vec{H}_1)=\vec{K}_f$ ➡ $\vec{H}_2^{II}-\vec{H}_1^{II}=\vec{K}_f\times\hat{n}$

- $E^{||}$ is always continuous across the boundary.
- $B^{\perp}$ is always continuous across the boundary.

In the absence of any free charge and free current, $\sigma_f=0;\ \vec{K}_f=0$

- $D^{\perp}$ is also continuous across the boundary.
- $H^{||}$ is also continuous across the boundary.

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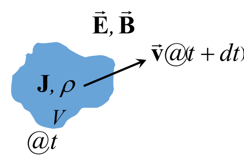
Poynting Theorem

How much work (dW) is done by the electromagnetic forces acting on these charges in dt?



$$w=\int\vec{\mathbf{F}}\cdot d\vec{\mathbf{l}}=\int q(\vec{\mathbf{E}}+\cancel{\vec{\mathbf{v}}\times\vec{\mathbf{B}})\cdot\vec{\mathbf{v}}}dt=\int q\vec{\mathbf{E}}\cdot\vec{\mathbf{v}}dt$$

$$q=\rho\,d\tau \quad \vec{\mathbf{J}}=\rho\,\vec{\mathbf{v}} \quad ➡ \quad \vec{\mathbf{v}}=\vec{\mathbf{J}}/\rho=\vec{\mathbf{J}}\,d\tau/q$$

➡ $w=\int\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\,d\tau\,dt$

Work done on all the charges in volume V: $\partial W=\int\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\,d\tau\,dt$

Work done per unit time on all the charges in volume V: $\dfrac{\partial W}{\partial t}=\int_V\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\,d\tau$ *power delivered per unit volume*

$$\vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\vec{\mathbf{J}}+\varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t} \quad ➡ \quad \vec{\mathbf{J}}=\frac{1}{\mu_0}\vec{\nabla}\times\vec{\mathbf{B}}-\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

➡ $\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}=\dfrac{1}{\mu_0}\vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}})-\varepsilon_0\vec{\mathbf{E}}\cdot\dfrac{\partial\vec{\mathbf{E}}}{\partial t}$

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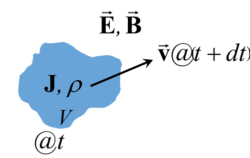
Poynting Theorem

How much work (dW) is done by the electromagnetic forces acting on these charges in dt?

Work done per unit time on all the charges in volume V:



$$\frac{\partial W}{\partial t}=\int_V\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\,d\tau \qquad \vec{\mathbf{E}}\cdot\vec{\mathbf{J}}=\frac{1}{\mu_0}\vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}})-\varepsilon_0\vec{\mathbf{E}}\cdot\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

$$\vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})=\vec{\mathbf{B}}\cdot(\vec{\nabla}\times\vec{\mathbf{E}})-\vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}}) \quad ➡ \quad \vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}})=\vec{\mathbf{B}}\cdot(\vec{\nabla}\times\vec{\mathbf{E}})-\vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})$$

$$=\vec{\mathbf{B}}\cdot\left(-\frac{\partial\vec{\mathbf{B}}}{\partial t}\right)-\vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})$$

➡ $$\frac{\partial W}{\partial t}=\int_V\vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\,d\tau=\frac{1}{\mu_0}\int_V\vec{\mathbf{B}}\cdot\left(-\frac{\partial\vec{\mathbf{B}}}{\partial t}\right)d\tau-\frac{1}{\mu_0}\int_V\vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})\,d\tau-\varepsilon_0\int_V\vec{\mathbf{E}}\cdot\left(\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)d\tau$$

$$\vec{\mathbf{B}}\cdot\left(\frac{\partial\vec{\mathbf{B}}}{\partial t}\right)=\frac{1}{2}\frac{\partial\mathbf{B}^2}{\partial t} \qquad \vec{\mathbf{E}}\cdot\left(\frac{\partial\vec{\mathbf{E}}}{\partial t}\right)=\frac{1}{2}\frac{\partial\mathbf{E}^2}{\partial t}$$

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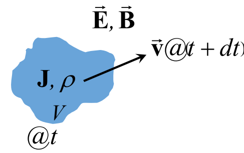
Poynting Theorem

How much work (dW) is done by the electromagnetic forces acting on these charges in dt?



Work done per unit time on all the charges in volume V:

$$\frac{\partial W}{\partial t} = \int_V \vec{\mathbf{E}}\cdot\vec{\mathbf{J}}\, d\tau \qquad \vec{\mathbf{E}}\cdot\vec{\mathbf{J}} = \frac{1}{\mu_0}\vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}}) - \varepsilon_0\vec{\mathbf{E}}\cdot\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

$$\vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}}) = \vec{\mathbf{B}}\cdot(\vec{\nabla}\times\vec{\mathbf{E}}) - \vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}}) \Rightarrow \vec{\mathbf{E}}\cdot(\vec{\nabla}\times\vec{\mathbf{B}}) = \vec{\mathbf{B}}\cdot(\vec{\nabla}\times\vec{\mathbf{E}}) - \vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}}) = \vec{\mathbf{B}}\cdot\left(-\frac{\partial\vec{\mathbf{B}}}{\partial t}\right) - \vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})$$

$$\Rightarrow \frac{\partial W}{\partial t} = -\frac{1}{2\mu_0}\int_V \frac{\partial\vec{\mathbf{B}}^2}{\partial t}d\tau - \frac{1}{\mu_0}\int_V \vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})\, d\tau - \frac{\varepsilon_0}{2}\int_V \frac{\partial\vec{\mathbf{E}}^2}{\partial t}d\tau$$

$$\vec{\mathbf{B}}\cdot\left(\frac{\partial\vec{\mathbf{B}}}{\partial t}\right) = \frac{1}{2}\frac{\partial\vec{\mathbf{B}}^2}{\partial t} \qquad \vec{\mathbf{E}}\cdot\left(\frac{\partial\vec{\mathbf{E}}}{\partial t}\right) = \frac{1}{2}\frac{\partial\vec{\mathbf{E}}^2}{\partial t}$$

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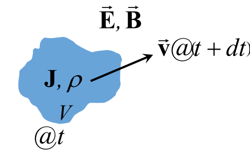
Poynting Theorem

How much work (dW) is done by the electromagnetic forces acting on these charges in dt?



Work done per unit time on all the charges in volume V:

$$\frac{\partial W}{\partial t} = -\frac{1}{2\mu_0}\int_V \frac{\partial\vec{\mathbf{B}}^2}{\partial t}d\tau - \frac{\varepsilon_0}{2}\int_V \frac{\partial\vec{\mathbf{E}}^2}{\partial t}d\tau - \frac{1}{\mu_0}\int_V \vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})\, d\tau$$

Total energy stored in the field

$$\boxed{\frac{\partial W}{\partial t} = -\frac{\partial}{\partial t}\int_V \frac{1}{2}\left(\varepsilon_0\vec{\mathbf{E}}^2 + \frac{1}{\mu_0}\vec{\mathbf{B}}^2\right)d\tau - \frac{1}{\mu_0}\oint_S (\vec{\mathbf{E}}\times\vec{\mathbf{B}})\cdot d\vec{\mathbf{a}}}$$

Poynting's theorem

Conservation of energy

$$\boxed{-\frac{\partial}{\partial t}\int_V \frac{1}{2}\left(\varepsilon_0\vec{\mathbf{E}}^2 + \frac{1}{\mu_0}\vec{\mathbf{B}}^2\right)d\tau = \frac{\partial W}{\partial t} + \frac{1}{\mu_0}\oint_S (\vec{\mathbf{E}}\times\vec{\mathbf{B}})\cdot d\vec{\mathbf{a}}}$$

Rate of decrease of total energy | Work done by the EM force | Remaining energy that flows out from V

$$\boxed{\vec{\mathbf{S}} = \frac{1}{\mu_0}(\vec{\mathbf{E}}\times\vec{\mathbf{B}})}$$

Poynting vector

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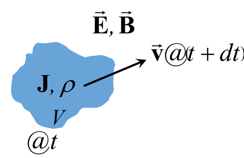
Poynting Theorem

How much work (dW) is done by the electromagnetic forces acting on these charges in dt?



Work done per unit time on all the charges in volume V:

$$\frac{\partial W}{\partial t} = -\frac{1}{2\mu_0}\int_V \frac{\partial\vec{\mathbf{B}}^2}{\partial t}d\tau - \frac{\varepsilon_0}{2}\int_V \frac{\partial\vec{\mathbf{E}}^2}{\partial t}d\tau - \frac{1}{\mu_0}\int_V \vec{\nabla}\cdot(\vec{\mathbf{E}}\times\vec{\mathbf{B}})\, d\tau$$

Total energy stored in the field

$$\boxed{\frac{\partial W}{\partial t} = -\frac{\partial}{\partial t}\int_V \frac{1}{2}\left(\varepsilon_0\vec{\mathbf{E}}^2 + \frac{1}{\mu_0}\vec{\mathbf{B}}^2\right)d\tau - \frac{1}{\mu_0}\oint_S (\vec{\mathbf{E}}\times\vec{\mathbf{B}})\cdot d\vec{\mathbf{a}}}$$

Poynting's theorem

Conservation of energy

$$\frac{\partial W}{\partial t} = -\frac{\partial U_{EM}}{\partial t} - \oint_S \vec{\mathbf{S}}\cdot d\vec{\mathbf{a}} \Rightarrow \frac{\partial}{\partial t}(W + U_{EM}) = -\oint_S \vec{\mathbf{S}}\cdot d\vec{\mathbf{a}}$$

$$\Rightarrow \frac{\partial}{\partial t}\int_V (U_{mec} + U_{EM})\, d\tau = -\int_V (\vec{\nabla}\cdot\vec{\mathbf{S}})d\tau \Rightarrow \boxed{\frac{\partial}{\partial t}(U_{mec} + U_{EM}) = -\vec{\nabla}\cdot\vec{\mathbf{S}}} \quad \boxed{\frac{\partial\rho}{\partial t} = -\vec{\nabla}\cdot\vec{\mathbf{J}}}$$

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Wave Theory

Maxwell Equation in Vacuum

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0} \qquad \vec{\nabla}\cdot\vec{\mathbf{B}} = 0 \qquad \vec{\nabla}\times\vec{\mathbf{E}} = -\frac{\partial\vec{\mathbf{B}}}{\partial t} \qquad \vec{\nabla}\times\vec{\mathbf{B}} = \mu_0\vec{\mathbf{J}} + \varepsilon_0\mu_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

For charge free, non-conducting medium $\rho = 0;\ \vec{\mathbf{J}} = 0$

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = 0 \qquad \vec{\nabla}\cdot\vec{\mathbf{B}} = 0 \qquad \vec{\nabla}\times\vec{\mathbf{E}} = -\frac{\partial\vec{\mathbf{B}}}{\partial t} \qquad \vec{\nabla}\times\vec{\mathbf{B}} = \mu_0\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{\mathbf{E}}) = \vec{\nabla}\cancel{(\vec{\nabla}\cdot\vec{\mathbf{E}})} - \nabla^2\vec{\mathbf{E}} = \vec{\nabla}\times\left(-\frac{\partial\vec{\mathbf{B}}}{\partial t}\right) = -\frac{\partial(\vec{\nabla}\times\vec{\mathbf{B}})}{\partial t} = -\varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}$$

$$\boxed{\nabla^2\vec{\mathbf{E}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} = 0}$$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{\mathbf{B}}) = \vec{\nabla}\cancel{(\vec{\nabla}\cdot\vec{\mathbf{B}})} - \nabla^2\vec{\mathbf{B}} = \mu_0\varepsilon_0\frac{\partial(\vec{\nabla}\times\vec{\mathbf{E}})}{\partial t} = -\varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}$$

$$\boxed{\nabla^2\vec{\mathbf{B}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2} = 0}$$

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Wave Theory

$$\nabla^2\vec{\mathbf{E}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} = 0$$

$$\nabla^2\vec{\mathbf{B}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2} = 0$$

$$\nabla^2\psi - \varepsilon_0\mu_0\frac{\partial^2\psi}{\partial t^2} = 0$$ **Scalar wave equation**

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Wave Theory

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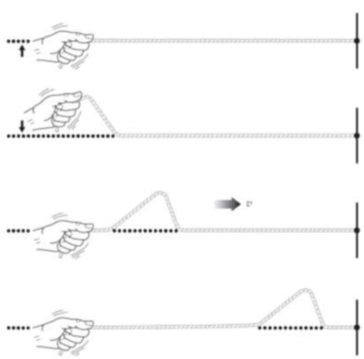
Wave Theory



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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \varepsilon_0\mu_0\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\psi - \varepsilon_0\mu_0\frac{\partial^2\psi}{\partial t^2} = 0}$$

Special case: One dimensional $\frac{\partial^2\psi}{\partial z^2} - \varepsilon_0\mu_0\frac{\partial^2\psi}{\partial t^2} = 0$

$\psi(z,t) = f(z,t)$ ➡ Progressive function (moving along z)

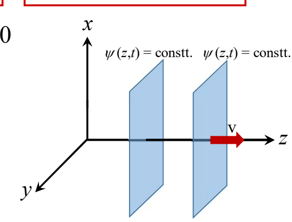
$\psi(z,t)|_{t=0} = f(z,0) = f(z)$ Shape/profile of ψ

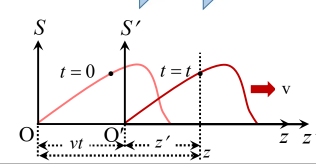
In reference frame, S': $\psi = f(z') = f(z - \mathrm{v}t)$

Let's assume $z - \mathrm{v}t = \omega$ $\psi(z,t) = f(z - \mathrm{v}t)$

$$\frac{\partial\psi}{\partial z} = \frac{\partial\psi}{\partial\omega}\frac{\partial\omega}{\partial z} = \frac{\partial\psi}{\partial\omega}$$

$$\frac{\partial^2\psi}{\partial z^2} = \frac{\partial}{\partial z}\frac{\partial\psi}{\partial\omega} = \frac{\partial}{\partial\omega}\frac{\partial\psi}{\partial\omega}\frac{\partial\omega}{\partial z} = \frac{\partial^2\psi}{\partial\omega^2}$$





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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \varepsilon_0\mu_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \varepsilon_0\mu_0 \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\psi - \varepsilon_0\mu_0 \frac{\partial^2 \psi}{\partial t^2} = 0}$$

Special case: One dimensional $\quad \frac{\partial^2 \psi}{\partial z^2} - \varepsilon_0\mu_0 \frac{\partial^2 \psi}{\partial t^2} = 0$

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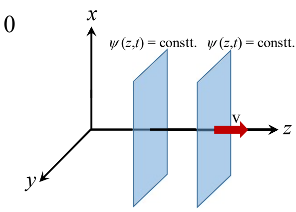
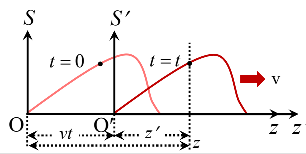
Let's assume $z - \mathrm{v}t = \omega$ $\qquad \psi(z,t) = f(z - \mathrm{v}t)$

$$\frac{\partial \psi}{\partial t} = \frac{\partial \psi}{\partial \omega}\frac{\partial \omega}{\partial t} = -\mathrm{v}\frac{\partial \psi}{\partial \omega}$$

$$\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial}{\partial t}\left(-\mathrm{v}\frac{\partial \psi}{\partial \omega}\right) = \frac{\partial}{\partial \omega}\left(-\mathrm{v}\frac{\partial \psi}{\partial \omega}\right)\frac{\partial \omega}{\partial t} = \mathrm{v}^2\frac{\partial^2 \psi}{\partial \omega^2}$$





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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \varepsilon_0\mu_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \varepsilon_0\mu_0 \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\psi - \varepsilon_0\mu_0 \frac{\partial^2 \psi}{\partial t^2} = 0}$$

Special case: One dimensional $\quad \frac{\partial^2 \psi}{\partial z^2} - \varepsilon_0\mu_0 \frac{\partial^2 \psi}{\partial t^2} = 0$

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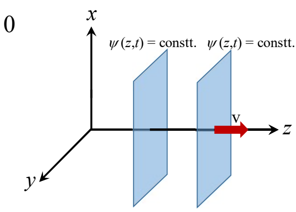
$\psi(z,t)\big|_{t=0} = f(z,0) = f(z)$ Shape/profile of ψ

In reference frame, S': $\psi = f(z') = f(z - \mathrm{v}t)$

Let's assume $z - \mathrm{v}t = \omega$ $\qquad \psi(z,t) = f(z - \mathrm{v}t)$

$$\frac{\partial^2 \psi}{\partial z^2} = \frac{\partial^2 \psi}{\partial \omega^2}; \quad \frac{\partial^2 \psi}{\partial t^2} = \mathrm{v}^2\frac{\partial^2 \psi}{\partial \omega^2}$$

$$\frac{\partial^2 \psi}{\partial \omega^2}\left(1 - \varepsilon_0\mu_0 \mathrm{v}^2\right) = 0 \;➡\; \boxed{\mathrm{v} = \frac{1}{\sqrt{\varepsilon_0\mu_0}} = c\,!}$$



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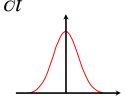
Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2} \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2} \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0}$$

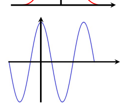
Special case: One dimensional $\quad \frac{\partial^2 \psi}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0$

$\psi(z,t) = f(z - \mathrm{v}t) \qquad \mathrm{v} = c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}$

$\psi(z,t) = f(z - \mathrm{v}t) = Ae^{-k(z-\mathrm{v}t)^2}$ Gaussian wave

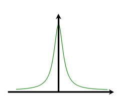


$\psi(z,t) = f(z - \mathrm{v}t) = A\sin k(z - vt)$ Sine wave



Harmonic Progressive wave

$\psi(z,t) = f(z - \mathrm{v}t) = \frac{A}{(z - vt)^2 + 1}$ Cusp wave



$\psi(z,t) \neq A\sin kz\cos(kvt)^2$

$\psi(z,t) \neq Ae^{-k(kz^2 + vt)}$

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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2} \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2} \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2} = 0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0}$$

Special case: One dimensional $\quad \frac{\partial^2 \psi}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0$ $\qquad \boxed{k^2 = \frac{\omega^2}{c^2}}$

$\psi(z,t) = f(z - \mathrm{v}t) \qquad \mathrm{v} = c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}$

$\psi(z,t) = Z(z)T(t)$

$$T\frac{d^2 Z}{dz^2} = \frac{1}{c^2}Z\frac{d^2 T}{dt^2} \quad \times\frac{1}{ZT}$$

$$\frac{1}{Z}\frac{d^2 Z}{dz^2} = \frac{1}{c^2}\frac{1}{T}\frac{d^2 T}{dt^2} = -k^2$$

$f(z) = f(t)$

$$\frac{1}{Z}\frac{d^2 Z}{dz^2} = -k^2 \;➡\; \frac{d^2 Z}{dz^2} + k^2 Z = 0 \;➡\; Z(z) = Ae^{ikz} + Be^{-ikz}$$

$$\frac{1}{c^2 T}\frac{d^2 T}{dt^2} = -k^2 \;➡\; \frac{d^2 T}{dt^2} + \omega^2 T = 0 \;➡\; T(t) = Ce^{i\omega t} + De^{-i\omega t}$$

$\psi(z,t): \quad e^{i(kz+\omega t)} \quad e^{i(kz-\omega t)} \quad e^{-i(kz-\omega t)} \quad e^{-i(kz+\omega t)} \qquad i = \sqrt{-1}$

$$\boxed{\psi(z,t) = \psi_0 e^{i(kz-\omega t)}}$$

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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}=0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}=0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0}$$

Special case: One dimensional $\frac{\partial^2\psi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0$

$\boxed{\psi(z,t)=\psi_0 e^{i(kz-\omega t)}}$ $\boxed{k^2=\frac{\omega^2}{c^2}}$

☞ $\psi_{physical} = \mathrm{Re}\{\psi_0 e^{i(kz-\omega t)}\} = \psi_0\cos(kz-\omega t)/\psi_0\sin(kz-\omega t)$

☞ ψ_0 itself is complex! $\psi_0=\psi_{0R}+i\psi_{0I}=\psi_{0a}e^{i\varepsilon}$ ➡ $\psi(z,t)=\tilde{\psi}_0 e^{i(kz-\omega t)}$

➡ $\psi(z,t)=\psi_{0a}e^{i(kz-\omega t+\varepsilon)}$ ➡ $\mathrm{Re}\{\psi(z,t)\}=\boxed{\psi_{0a}\cos(kz-\omega t+\varepsilon)=\psi(z,t)}$

☞ ψ_0 is periodic in space and time! $\cos(kz-\omega t+\varepsilon+2\pi)=\cos(kz-\omega t+\varepsilon)$

LHS: $\cos\{kz+2\pi-\omega t+\varepsilon\}=\cos\{k(z+2\pi/k)-\omega t+\varepsilon\}=\cos(kz-\omega t+\varepsilon)$

➡ Spatial periodic: $\boxed{\lambda=2\pi/k}$ Wavelength

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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}=0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}=0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0}$$

Special case: One dimensional $\frac{\partial^2\psi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0$

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➡ $\psi(z,t)=\psi_{0a}e^{i(kz-\omega t+\varepsilon)}$ ➡ $\mathrm{Re}\{\psi(z,t)\}=\boxed{\psi_{0a}\cos(kz-\omega t+\varepsilon)=\psi(z,t)}$

☞ ψ_0 is periodic in space and time! $\cos(kz-\omega t+\varepsilon+2\pi)=\cos(kz-\omega t+\varepsilon)$

LHS: $\cos\{kz-\omega t+2\pi+\varepsilon\}=\cos\{kz-\omega(t-2\pi/\omega)+\varepsilon\}=\cos(kz-\omega t+\varepsilon)$

➡ Time periodic: $\boxed{T=2\pi/\omega}$ $\boxed{\nu=1/T}=\omega/2\pi$ ➡ $\boxed{\omega=2\pi\nu}$

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Wave Theory

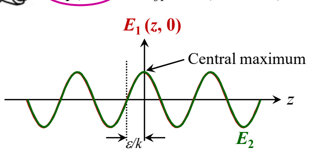
$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}=0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}=0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0}$$

Special case: One dimensional $\frac{\partial^2\psi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0$

$\boxed{\psi(z,t)=\psi_0\cos(kz-\omega t+\varepsilon)}$

☞ $\lambda=2\pi/k;\ k=\omega/c;\ \omega=2\pi\nu$ ➡ $\omega=2\pi\nu=kc$ ➡ $\cancel{2\pi}\nu=\frac{\cancel{2\pi}}{\lambda}c$ ➡ $\boxed{c=\nu\lambda}$

☞ $\vec{\mathbf{E}}_1(z,t)=\mathbf{E}_{01}\cos(kz-\omega t)$ $\vec{\mathbf{E}}_2(z,t)=\mathbf{E}_{02}\cos(kz-\omega t+\varepsilon)$



at $z=\frac{\omega}{k}t-\frac{\varepsilon}{k}$; $|\vec{\mathbf{E}}_2(z,t)|=\mathrm{E}_{02}$: Central maximum

For $\varepsilon=0$ at $t=0$, central maximum passes through the origin (appears at $z=0$).

For $\varepsilon\neq 0$ at $t=0$, central maximum appears at $z=-\varepsilon/k$.

For +ve ε, E_2 lags behind E_1

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Wave Theory

$$\boxed{\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}=0} \quad \boxed{\nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}=0} \quad \boxed{\nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0}$$

Special case: One dimensional $\frac{\partial^2\psi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0$

$\vec{\mathbf{E}}=\vec{\mathbf{E}}_0 e^{i(kz-\omega t)}$ $\vec{\mathbf{B}}=\vec{\mathbf{B}}_0 e^{i(kz-\omega t)}$

☞ $\vec{\nabla}\cdot\vec{\mathbf{E}}=0$ ➡ $\cancel{\frac{\partial E_x}{\partial x}}+\cancel{\frac{\partial E_y}{\partial y}}+\frac{\partial E_z}{\partial z}=0$ ➡ $ikE_z=0$ ➡ $k\,\hat{z}\cdot\vec{\mathbf{E}}=0$

☞ $\vec{\nabla}\cdot\vec{\mathbf{B}}=0$ ➡ $k\,\hat{z}\cdot\vec{\mathbf{B}}=0$

☞ $\vec{\nabla}\times\vec{\mathbf{E}}=-\frac{\partial\vec{\mathbf{B}}}{\partial t}$ ➡ $\begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ E_x & E_y & E_z\end{vmatrix}=-i\omega\vec{\mathbf{B}}$ ➡ $\hat{x}\left(-\frac{\partial E_y}{\partial z}\right)-\hat{y}\left(-\frac{\partial E_x}{\partial z}\right)=i\omega\vec{\mathbf{B}}$

➡ $-ik\hat{x}E_y+ik\hat{y}E_x=i\omega\vec{\mathbf{B}}$

☞ $\vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\varepsilon_0\frac{\partial\vec{\mathbf{E}}}{\partial t}$ ➡ $k\hat{z}\times\vec{\mathbf{B}}=-\mu_0\varepsilon_0\omega\vec{\mathbf{E}}$

➡ $k(\hat{y}E_x-\hat{x}E_y)=\omega\vec{\mathbf{B}}$

➡ $k\hat{z}\times\vec{\mathbf{E}}=\omega\vec{\mathbf{B}}$

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Wave Theory

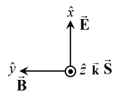
$$\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} = 0 \qquad \nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2} = 0 \qquad \nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = 0$$

Special case: One dimensional $\quad \frac{\partial^2\psi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = 0$

$\vec{\mathbf{E}} = \vec{\mathbf{E}}_0 e^{i(kz-\omega t)} \qquad \vec{\mathbf{B}} = \vec{\mathbf{B}}_0 e^{i(kz-\omega t)}$

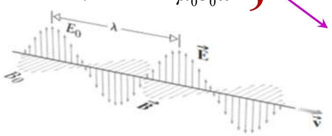
$k^2 = \frac{\omega^2}{c^2}$

$$\left.\begin{aligned} k\,\hat{z}\cdot\vec{\mathbf{E}} &= 0; \\ k\,\hat{z}\cdot\vec{\mathbf{B}} &= 0; \end{aligned}\right\} \vec{\mathbf{E}} \perp \vec{\mathbf{k}};\ \vec{\mathbf{B}} \perp \vec{\mathbf{k}}$$



$$\left.\begin{aligned} k\hat{z}\times\vec{\mathbf{E}} &= \omega\vec{\mathbf{B}} \\ k\hat{z}\times\vec{\mathbf{B}} &= -\mu_0\varepsilon_0\omega\vec{\mathbf{E}} \end{aligned}\right\} \vec{\mathbf{B}} \perp \vec{\mathbf{k}} \perp \vec{\mathbf{E}}$$

$$\boxed{\vec{\mathbf{B}} = \frac{k}{\omega}\hat{z}\times\vec{\mathbf{E}} = \frac{1}{c}\hat{z}\times\vec{\mathbf{E}}}$$



$$\Rightarrow \boxed{\left|\vec{\mathbf{B}}\right| = \frac{1}{c}\left|\vec{\mathbf{E}}\right|}$$

can k be zero?

yes, why not?

$k = \omega / c;$

Oh, yup, but that will be ***static case*** *and I am* ***not interested*** *in it anymore!*

hmm…for non-static case, $k \neq 0, \omega \neq 0$

$\mathbf{E}_z = 0 = \mathbf{B}_z$

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Wave Theory

$$\nabla^2\vec{\mathbf{E}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} = 0 \qquad \nabla^2\vec{\mathbf{B}} - \frac{1}{c^2}\frac{\partial^2\vec{\mathbf{B}}}{\partial t^2} = 0 \qquad \nabla^2\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = 0$$

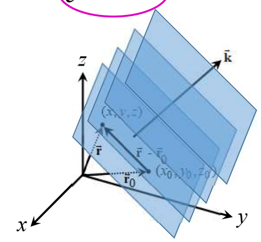
General case: Three dimensional $\quad \psi(x,y,z,t):\ e^{i(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}}+\omega t)}\ \ e^{i(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}}-\omega t)}$

$\vec{\mathbf{k}} = k_x\hat{\mathbf{x}} + k_y\hat{\mathbf{y}} + k_z\hat{\mathbf{z}}$

$\vec{\mathbf{r}}_0 = x_0\hat{\mathbf{x}} + y_0\hat{\mathbf{y}} + z_0\hat{\mathbf{z}} \qquad \vec{\mathbf{r}} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$

$\vec{\mathbf{k}}\cdot(\vec{\mathbf{r}} - \vec{\mathbf{r}}_0) = 0 \Rightarrow \vec{\mathbf{k}}\cdot\vec{\mathbf{r}} = \text{constant}$

Equation of a plane $\perp \vec{\mathbf{k}}$



- Solution of wave equation is in the form of plane wave
  $\vec{\mathbf{E}}(\vec{\mathbf{r}},t) = \vec{\mathbf{E}}_0 e^{i(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}}-\omega t)} \qquad \vec{\mathbf{B}}(\vec{\mathbf{r}},t) = \vec{\mathbf{B}}_0 e^{i(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}}-\omega t)}$
- $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ are in phase.
- $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ fields of the wave travel in vacuum at a speed: $\text{v} = c = \frac{1}{\sqrt{\varepsilon_0\mu_0}} = 3\times10^8$ m/s

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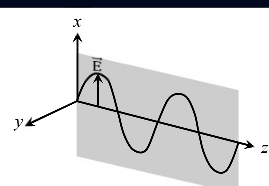
Polarization

Linear (Plane) Polarization ($\mathscr{P}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$



ε: Relative phase difference between $\vec{\mathbf{E}}_{\mathbf{x}}$ and $\vec{\mathbf{E}}_{\mathbf{y}}$

☞ (a) $\varepsilon > 0$: E_y lags E_x; (b) $\varepsilon < 0$: E_y leads E_x

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Polarization

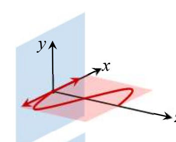
Linear (Plane) Polarization ($\mathscr{P}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

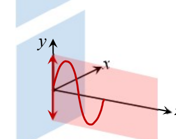
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

Resulting disturbance: $\vec{\mathbf{E}} = \vec{\mathbf{E}}_{\mathbf{x}}(z,t) + \vec{\mathbf{E}}_{\mathbf{y}}(z,t)$

- **A. Horizontally polarized light:**
  $\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t);$
  $\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = 0$



- **B. Vertically polarized light**
  $\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = 0;$
  $\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$



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Polarization

Linear (Plane) Polarization ($\mathscr{P}$-state)

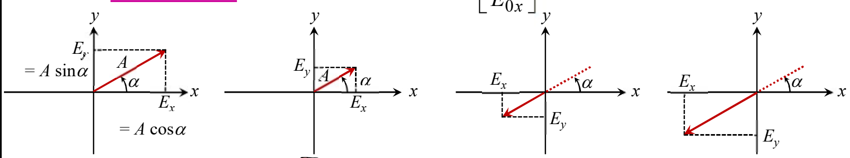
$\vec{\mathbf{E}}_\mathbf{x}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

$\vec{\mathbf{E}}_\mathbf{y}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

Resulting disturbance: $\vec{\mathbf{E}} = \vec{\mathbf{E}}_\mathbf{x}(z,t) + \vec{\mathbf{E}}_\mathbf{y}(z,t)$

C. Linearly polarized light $\varepsilon = 2m\pi,\quad m = 0,\pm1,\pm2,\pm3,\cdots$ waves are in phase

$$\vec{\mathbf{E}} = \left[\hat{\mathbf{x}}\,\mathrm{E}_{0x} + \hat{\mathbf{y}}\,\mathrm{E}_{0y}\right]\cos(kz-\omega t) \qquad \alpha = \tan^{-1}\left[\frac{E_{0y}}{E_{0x}}\right]$$



Linearly polarized light (at $+\alpha$)

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Polarization

Linear (Plane) Polarization ($\mathscr{P}$-state)

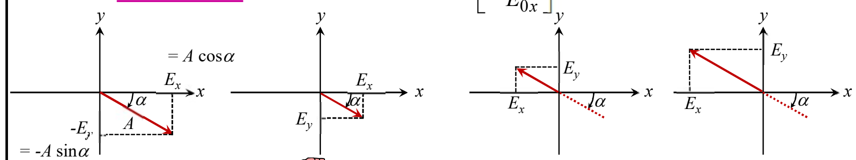
$\vec{\mathbf{E}}_\mathbf{x}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

$\vec{\mathbf{E}}_\mathbf{y}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

Resulting disturbance: $\vec{\mathbf{E}} = \vec{\mathbf{E}}_\mathbf{x}(z,t) + \vec{\mathbf{E}}_\mathbf{y}(z,t)$

D. Linearly polarized light $\varepsilon = (2m+1)\pi,\quad m = 0,\pm1,\pm2,\pm3,\cdots$ waves are 180° out of phase

$$\vec{\mathbf{E}} = \left[\hat{\mathbf{x}}\,\mathrm{E}_{0x} - \hat{\mathbf{y}}\,\mathrm{E}_{0x}\right]\cos(kz-\omega t) \qquad \alpha = \tan^{-1}\left[-\frac{E_{0y}}{E_{0x}}\right]$$



Linearly polarized light (at $-\alpha$)

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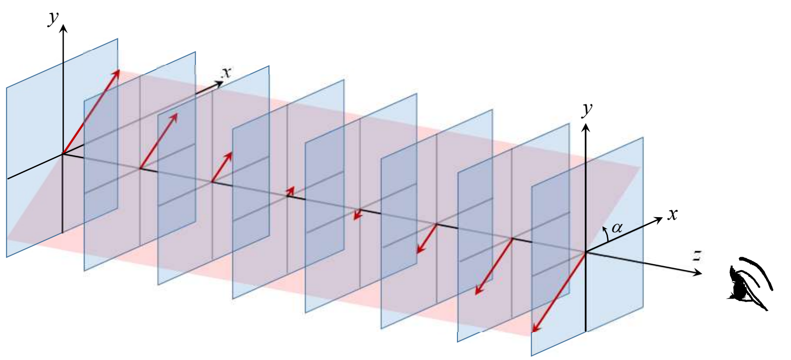
Polarization

Linear (Plane) Polarization ($\mathscr{P}$-state)



Linearly polarized light (at $+\alpha$)

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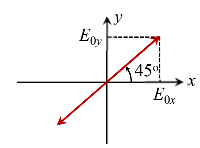
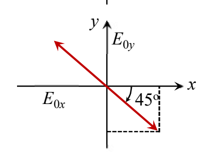
Polarization

Linear (Plane) Polarization ($\mathscr{P}$-state)

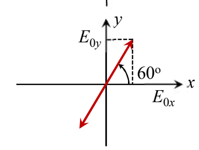
Example 1: Linearly polarized light oriented at 45°.



Example 2: Linearly polarized light oriented at -45°.



Example 3: Linearly polarized light oriented at 60°.



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Polarization

Circular Polarization ($\mathscr{R}$-state)

$\vec{\mathbf{E}}_x(z,t) = \hat{\mathbf{x}}\,E_{0x}\cos(kz-\omega t)$

$\vec{\mathbf{E}}_y(z,t) = \hat{\mathbf{y}}\,E_{0y}\cos(kz-\omega t+\varepsilon)$

$\varepsilon = -\frac{\pi}{2}, -\frac{5\pi}{2}, -\frac{9\pi}{2}, -\frac{13\pi}{2}, \cdots$

$\frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \frac{15\pi}{2}, \cdots$

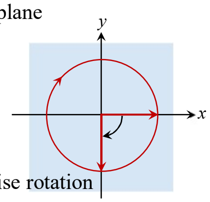
E. Case 1: $E_{0x} = E_{0y}$; $\varepsilon = 2m\pi - \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_x(z,t) &= \hat{\mathbf{x}}\,E_0\cos(kz-\omega t)\\ \vec{\mathbf{E}}_y(z,t) &= \hat{\mathbf{y}}\,E_0\sin(kz-\omega t)\end{aligned}\right\}$$

$\cos(kz-\omega t+2m\pi-\pi/2) = \cos(kz-\omega t-\pi/2)$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = \hat{\mathbf{x}}\,E_0\cos\omega t\\ \vec{\mathbf{E}}_y = -\hat{\mathbf{y}}\,E_0\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = \hat{\mathbf{x}}\,E_0\\ \vec{\mathbf{E}}_y = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = 0\\ \vec{\mathbf{E}}_y = -\hat{\mathbf{y}}\,E_0\end{cases}$



Clock-wise rotation

Right Circular Polarization

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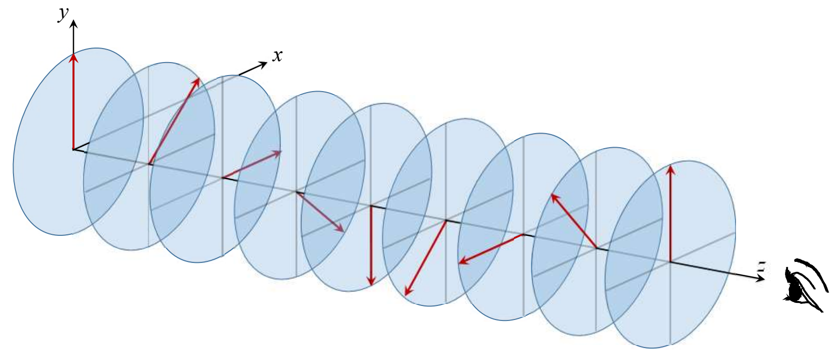
Polarization

Circular Polarization ($\mathscr{R}$-state)



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Polarization

Circular Polarization ($\mathscr{L}$-state)

$\vec{\mathbf{E}}_x(z,t) = \hat{\mathbf{x}}\,E_{0x}\cos(kz-\omega t)$

$\vec{\mathbf{E}}_y(z,t) = \hat{\mathbf{y}}\,E_{0y}\cos(kz-\omega t+\varepsilon)$

$\varepsilon = -\frac{3\pi}{2}, -\frac{7\pi}{2}, -\frac{11\pi}{2}, -\frac{15\pi}{2}, \cdots$

$\frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \cdots$

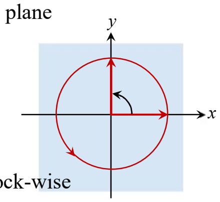
E. Case 2: $E_{0x} = E_{0y}$; $\varepsilon = 2m\pi + \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_x(z,t) &= \hat{\mathbf{x}}\,E_0\cos(kz-\omega t)\\ \vec{\mathbf{E}}_y(z,t) &= -\hat{\mathbf{y}}\,E_0\sin(kz-\omega t)\end{aligned}\right\}$$

$\cos(kz-\omega t+2m\pi+\pi/2) = \cos(kz-\omega t+\pi/2)$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = \hat{\mathbf{x}}\,E_0\cos\omega t\\ \vec{\mathbf{E}}_y = \hat{\mathbf{y}}\,E_0\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = \hat{\mathbf{x}}\,E_0\\ \vec{\mathbf{E}}_y = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad \begin{cases}\vec{\mathbf{E}}_x = 0\\ \vec{\mathbf{E}}_y = \hat{\mathbf{y}}\,E_0\end{cases}$



Anti-clock-wise rotation

Left Circular Polarization

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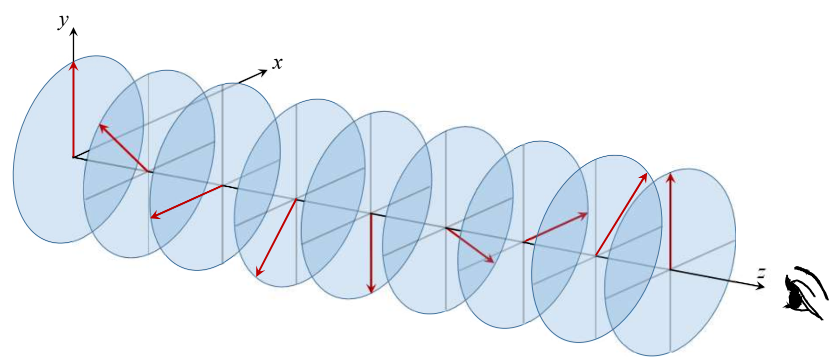
Polarization

Circular Polarization ($\mathscr{L}$-state)



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Polarization

Elliptical Polarization ($\mathscr{E}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

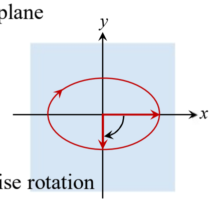
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

E. Case 1: $E_{0x} \neq E_{0y}\,;\ \varepsilon = 2m\pi - \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_{\mathbf{x}}(z,t) &= \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)\\ \vec{\mathbf{E}}_{\mathbf{y}}(z,t) &= \hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin(kz-\omega t)\end{aligned}\right\}\quad \cos(kz-\omega t+2m\pi-\pi/2) = \cos(kz-\omega t-\pi/2)$$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos\omega t\\ \vec{\mathbf{E}}_{\mathbf{y}} = -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\\ \vec{\mathbf{E}}_{\mathbf{y}} = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = 0\\ \vec{\mathbf{E}}_{\mathbf{y}} = -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\quad (\mathrm{E}_{0y} < \mathrm{E}_{0x})\end{cases}$



Clock-wise rotation

Right Elliptical Polarization

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Polarization

Elliptical Polarization ($\mathscr{E}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

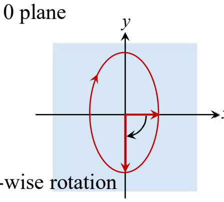
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

E. Case 1: $E_{0x} \neq E_{0y}\,;\ \varepsilon = 2m\pi - \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_{\mathbf{x}}(z,t) &= \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)\\ \vec{\mathbf{E}}_{\mathbf{y}}(z,t) &= \hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin(kz-\omega t)\end{aligned}\right\}\quad \cos(kz-\omega t+2m\pi-\pi/2) = \cos(kz-\omega t-\pi/2)$$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos\omega t\\ \vec{\mathbf{E}}_{\mathbf{y}} = -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\\ \vec{\mathbf{E}}_{\mathbf{y}} = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = 0\\ \vec{\mathbf{E}}_{\mathbf{y}} = -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\quad (\mathrm{E}_{0y} > \mathrm{E}_{0x})\end{cases}$



Clock-wise rotation

Right Elliptical Polarization

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Polarization

Elliptical Polarization ($\mathscr{E}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

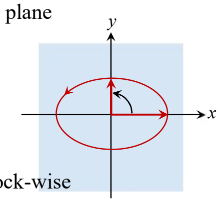
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

E. Case 2: $E_{0x} \neq E_{0y}\,;\ \varepsilon = 2m\pi + \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_{\mathbf{x}}(z,t) &= \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)\\ \vec{\mathbf{E}}_{\mathbf{y}}(z,t) &= -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin(kz-\omega t)\end{aligned}\right\}\quad \cos(kz-\omega t+2m\pi+\pi/2) = \cos(kz-\omega t+\pi/2)$$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos\omega t\\ \vec{\mathbf{E}}_{\mathbf{y}} = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\\ \vec{\mathbf{E}}_{\mathbf{y}} = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = 0\\ \vec{\mathbf{E}}_{\mathbf{y}} = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\quad (\mathrm{E}_{0y} < \mathrm{E}_{0x})\end{cases}$



Anti-clock-wise rotation

Left Elliptical Polarization

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Polarization

Elliptical Polarization ($\mathscr{E}$-state)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)$

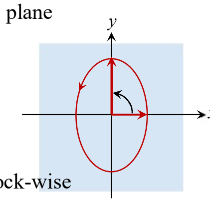
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$

E. Case 2: $E_{0x} \neq E_{0y}\,;\ \varepsilon = 2m\pi + \pi/2 \quad (m = 0, \pm1, \pm2, \pm3, \cdots)$

$$\left.\begin{aligned}\vec{\mathbf{E}}_{\mathbf{x}}(z,t) &= \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t)\\ \vec{\mathbf{E}}_{\mathbf{y}}(z,t) &= -\hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin(kz-\omega t)\end{aligned}\right\}\quad \cos(kz-\omega t+2m\pi+\pi/2) = \cos(kz-\omega t+\pi/2)$$

Analyze these fields at, say, $z = 0$ plane

- At $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos\omega t\\ \vec{\mathbf{E}}_{\mathbf{y}} = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\sin\omega t\end{cases}$
- At $t = 0$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\\ \vec{\mathbf{E}}_{\mathbf{y}} = 0\end{cases}$
- At $t = \pi/2\omega$ on $z = 0$ $\quad\begin{cases}\vec{\mathbf{E}}_{\mathbf{x}} = 0\\ \vec{\mathbf{E}}_{\mathbf{y}} = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\quad (\mathrm{E}_{0y} > \mathrm{E}_{0x})\end{cases}$



Anti-clock-wise rotation

Left Elliptical Polarization

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Polarization

Polarization (General case)

$\vec{\mathbf{E}}_{\mathbf{x}}(z,t) = \hat{\mathbf{x}}\,\mathrm{E}_{0x}\cos(kz-\omega t) \Rightarrow \mathrm{E}_x = \mathrm{E}_{0x}\cos(kz-\omega t)$ $\qquad E_{0x} \neq E_{0y}$;

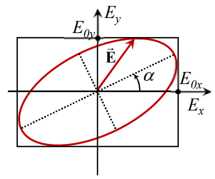
$\vec{\mathbf{E}}_{\mathbf{y}}(z,t) = \hat{\mathbf{y}}\,\mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon) \Rightarrow \mathrm{E}_y = \mathrm{E}_{0y}\cos(kz-\omega t+\varepsilon)$ $\qquad \varepsilon \neq 0, \pm\pi, \pm 2\pi, \pm 3\pi, \cdots$

$$\frac{\mathrm{E}_y}{\mathrm{E}_{0y}} = \cos(kz-\omega t)\cos\varepsilon - \sin(kz-\omega t)\sin\varepsilon \qquad \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \cdots$$

$$= \frac{\mathrm{E}_x}{\mathrm{E}_{0x}}\cos\varepsilon - \left(1-\frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2}\right)^{1/2}\sin\varepsilon \Rightarrow \left(\frac{\mathrm{E}_y}{\mathrm{E}_{0y}} - \frac{\mathrm{E}_x}{\mathrm{E}_{0x}}\cos\varepsilon\right)^2 = \left(1-\frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2}\right)\sin^2\varepsilon$$

$$\frac{\mathrm{E}_y^2}{\mathrm{E}_{0y}^2} + \frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2}\cos^2\varepsilon - \frac{2\mathrm{E}_x\mathrm{E}_y}{\mathrm{E}_{0x}\mathrm{E}_{0y}}\cos\varepsilon = \sin^2\varepsilon - \frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2}\sin^2\varepsilon$$

$$\boxed{\frac{\mathrm{E}_y^2}{\mathrm{E}_{0y}^2} + \frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2} - \frac{2\mathrm{E}_x\mathrm{E}_y}{\mathrm{E}_{0x}\mathrm{E}_{0y}}\cos\varepsilon = \sin^2\varepsilon} \qquad \boxed{\tan 2\alpha = \frac{2\mathrm{E}_{0x}\mathrm{E}_{0y}}{\mathrm{E}_{0x}^2-\mathrm{E}_{0y}^2}\cos\varepsilon}$$



Equation of ellipse making an angle α from E_x coordinate axis

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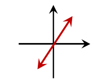
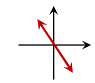
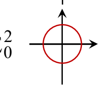
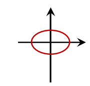
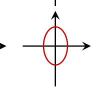
Polarization

Polarization (General case)

$$\boxed{\frac{\mathrm{E}_y^2}{\mathrm{E}_{0y}^2} + \frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2} - \frac{2\mathrm{E}_x\mathrm{E}_y}{\mathrm{E}_{0x}\mathrm{E}_{0y}}\cos\varepsilon = \sin^2\varepsilon} \qquad \boxed{\tan 2\alpha = \frac{2\mathrm{E}_{0x}\mathrm{E}_{0y}}{\mathrm{E}_{0x}^2-\mathrm{E}_{0y}^2}\cos\varepsilon}$$

- **Special case A:** $\varepsilon = 0, \pm 2\pi, \pm 4\pi, \cdots$ $\quad \left(\frac{\mathrm{E}_y}{\mathrm{E}_{0y}} - \frac{\mathrm{E}_x}{\mathrm{E}_{0x}}\right)^2 = 0 \Rightarrow \mathrm{E}_y = \frac{\mathrm{E}_{0y}}{\mathrm{E}_{0x}}\mathrm{E}_x$ 
- **Special case B:** $\varepsilon = \pm\pi, \pm 3\pi, \pm 5\pi \cdots$ $\quad \left(\frac{\mathrm{E}_y}{\mathrm{E}_{0y}} + \frac{\mathrm{E}_x}{\mathrm{E}_{0x}}\right)^2 = 0 \Rightarrow \mathrm{E}_y = -\frac{\mathrm{E}_{0y}}{\mathrm{E}_{0x}}\mathrm{E}_x$ 
- **Special case C:** $\mathrm{E}_{0x} = \mathrm{E}_{0y}$; $\varepsilon = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$ $\quad \frac{\mathrm{E}_y^2}{\mathrm{E}_0^2} + \frac{\mathrm{E}_x^2}{\mathrm{E}_0^2} = 1 \Rightarrow \mathrm{E}_x^2 + \mathrm{E}_y^2 = \mathrm{E}_0^2$ 
- **Special case D:** $\mathrm{E}_{0x} \neq \mathrm{E}_{0y}$; $\varepsilon = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$ $\quad \frac{\mathrm{E}_y^2}{\mathrm{E}_{0y}^2} + \frac{\mathrm{E}_x^2}{\mathrm{E}_{0x}^2} = 1$  

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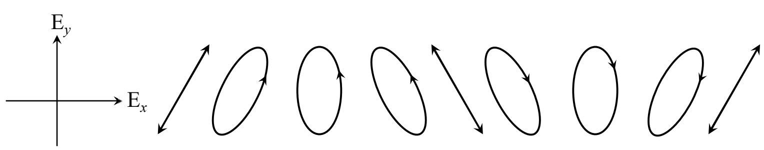
Polarization

Polarization (General case)



$\vec{\mathbf{E}}_{\mathbf{x}}$ leads $\vec{\mathbf{E}}_{\mathbf{y}}$ by $\quad 0 \quad \pi/4 \quad \pi/2 \quad 3\pi/4 \quad \pi \quad 5\pi/4 \quad 3\pi/2 \quad 7\pi/4 \quad 2\pi \quad (=\varepsilon)$

Left handed ($\pi/4$ to $3\pi/4$) $\qquad$ **Right handed** ($5\pi/4$ to $7\pi/4$)

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